PASTA Threat Model Analysis: Sneaker Company Mobile Application

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Stages	Sneaker company
I. Define business and security objectives	• Process secure online payments for buying and selling sneakers. • Protect users' personal and payment information to maintain privacy and trust. • Comply with PCI DSS requirements and provide secure account management.
II. Define the technical scope	Technologies used by the application: • *Application programming interface (API)* • *Public key infrastructure (PKI)* • *SHA-256* • *SQL* I would prioritize the API because it connects the mobile app with the database, authentication system, and payment services. It handles sensitive customer information and communication between multiple systems, making it a common target for attackers. Securing the API helps reduce the application's overall attack surface.
III. Decompose application	Sample data flow diagram
IV. Threat analysis	**2 types of threats** in the PASTA worksheet that are risks to the information being handled by the application. External threat: SQL injection attacks targeting the application's database. Internal threat: An employee misusing authorized access to customer information.
V. Vulnerability analysis	**2 vulnerabilities** in the PASTA worksheet that could be exploited. Code vulnerability: Poor input validation could allow attackers to enter harmful data and perform SQL injection attacks. Network vulnerability: Weak security during data transfer could allow attackers to steal sensitive user information .

VI. Attack modeling	Sample attack tree diagram
VII. Risk analysis and impact	**security controls** ● Implement multi-factor authentication (MFA). ● Hash passwords using SHA-256 and encrypt sensitive data. ● Use prepared statements and input validation to prevent SQL injection. ● Apply the Principle of Least Privilege (PoLP) and monitor system logs.
